Chapter 1, Problem 37CP

Problem

A total of 2 MJ are delivered to an automobile battery (assume 12 V) giving it an additional charge. How much is that additional charge? Express your answer in ampere-hours.

Step-by-step solution

Step 1 of 2

The power delivered is, 2 MJ.

The voltage of automobile battery is, 12 V.

Write the formula for charge.

$$q = it \text{ A.sec}$$

It is known that,

$$1 \text{ sec} = \frac{1}{60 \times 60} \text{ hr}$$
$$1 \text{ Coulomb} = 1 \text{ A.sec}$$

Step 2 of 2

Write the formula for power delivered.

$$\begin{aligned} P_d &= Vi \\ &= V\left(\frac{q}{t}\right) \end{aligned}$$

From the equation, the additional charge is,

$$q = \frac{P_d t}{V}$$

Substitute 12 V for V and 2 MJ for $P_d t$.

$$\begin{aligned} q &= \frac{2 \times 10^6}{12} \\ &= 166.67 \times 10^3 \text{ C} \\ &= 166.67 \times 10^3 \text{ A-sec} \\ &= \frac{166.67 \times 10^3}{60 \times 60} \text{ A-hr} \end{aligned}$$

$$= 46.29 \text{ A-hr}$$

Therefore, the additional charge, q is $\boxed{46.29 \text{ A-hrs}}$.